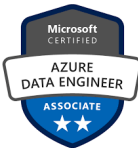


KAJAL PURI

Data Engineer

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PROFESSIONAL SUMMARY

Detail-oriented and cloud-native Data Engineer with 6 years of experience designing and building scalable data platforms across Azure and AWS ecosystems. Proficient in developing distributed ELT pipelines, modeling cloud data warehouses, and orchestrating batch and real-time workflows using tools like Databricks, Snowflake, Airflow, Kafka, and dbt. Strong in implementing Delta Lake architectures, Data Vault/Star schema modeling, and enabling secure, analytics-ready datasets across healthcare, pharma, and automotive domains. Hands-on with Spark (Scala/PySpark) and SQL for data transformation, and skilled in deploying infrastructure through Terraform, GitHub Actions, and Azure DevOps. Experienced in delivering curated data for ML pipelines, building reusable frameworks, and ensuring governance and data quality using Great Expectations and modern CI/CD practices.

PROFESSIONAL WORK EXPERIENCE

Data Engineer | Optum | Irving, Texas *Jun 2023 – Present*

- Designed and deployed ELT pipelines using Azure Data Factory, Databricks (Scala/PySpark), and Delta Lake, enabling scalable ingestion and transformation of healthcare datasets.
- Built multi-layered Delta Lake architecture (bronze, silver, gold) on ADLS Gen2, supporting modular pipeline development with optimized read/write performance.
- Developed notebook-based transformation logic in Databricks using Spark SQL and Scala, applying reusable business rules version-controlled via Git.
- Enabled real-time data ingestion of clinical event streams through Azure Event Hub and Azure Functions, integrating EHR system feeds with lakehouse storage.
- Modeled regulatory-compliant data marts in Azure Synapse Analytics using Star schema and Data Vault, tailored for population health and provider performance reporting.
- Created and maintained dbt models with macros and Jinja templates, improving code reusability and maintaining lineage through documentation.
- Embedded Great Expectations validation checks into transformation pipelines, ensuring schema conformity, null thresholds, and field-level accuracy.
- Managed secure access controls via Azure Key Vault and RBAC, enforcing data privacy policies aligned with HIPAA and HITRUST guidelines.
- Automated infrastructure provisioning with Terraform, supporting repeatable environment creation and drift detection across dev, QA, and prod tiers.
- Integrated Azure DevOps for CI/CD pipeline execution, including automated testing, linting, deployment, and artifact versioning.
- Supported curated datasets for Power BI dashboards, exposing Synapse views through defined semantic models for self-service analytics.
- Collaborated in Agile sprint planning and daily standups with data stewards, analysts, and governance teams to prioritize backlog items and resolve data quality issues.

Technologies Used: Azure Data Factory, Azure Databricks, Synapse Analytics, Delta Lake, Azure Data Lake Gen2, Event Hub, Azure Functions, dbt, Great Expectations, Terraform, Azure DevOps, Git, Spark SQL, Scala, Power BI, Key Vault, Agile.

Data Engineer | Infosys /Ford Motors | Dearborn, Michigan *Aug 2022 - May 2023*

- Built distributed ingestion pipelines using AWS Glue, Scala-based Spark jobs, and S3, supporting the processing of vehicle telemetry, diagnostics, and event logs.
- Integrated real-time data streams from connected vehicles using Kafka and IBM MQ, then routed messages to S3 buckets for structured processing.
- Developed Spark-based data transformations on EMR, applying schema evolution and mapping logic for fault codes, firmware updates, and ignition events.
- Used AWS Step Functions to orchestrate batch and micro-batch workflows, coordinating Lambda triggers, data validation, and alerting steps.
- Designed and optimized Redshift schemas with materialized views and partitioning to support warranty intelligence, OTA tracking, and part failure reporting.
- Built containerized services in Docker, deployed to Amazon EKS, with CI/CD pipelines managed via GitHub Actions for version-controlled delivery.

- Managed secure inter-service communication using IAM roles, Secrets Manager, and token-based authentication for REST API integrations.
- Supported time-series pipeline development using Kinesis Firehose, enabling continuous data streaming to Redshift with retry and buffer logic.
- Assisted QA team by delivering enriched event datasets for automated validation using Postman and PyTest scripts.
- Documented API endpoints, data flows, and transformation rules in Confluence, ensuring transparency for downstream analytics and support teams.

Environment: AWS Glue, Amazon Redshift, Apache Spark (Scala), EMR, S3, Kafka, IBM MQ, AWS Step Functions, Lambda, Amazon EKS, Docker, GitHub Actions, Kinesis Firehose, IAM, Secrets Manager, REST APIs, Confluence, PyTest.

Junior Data Engineer | Merck & Co. | Rahway, New Jersey *Jun 2019– Jul 2022*

- Designed ingestion pipelines using Apache NiFi and AWS Glue to extract structured datasets from internal enterprise systems and third-party vendors.
- Developed batch processing workflows in PySpark on EMR, applying data cleaning, schema normalization, and type casting for ingestion into the lake.
- Built dbt transformation models layered on Snowflake, incorporating reusable macros, snapshot logic, and dependency management.
- Used Apache Airflow to schedule and monitor workflows, integrating with Slack and email for alerting and success/failure notifications.
- Modeled warehousing structures in Snowflake using Data Vault and Star schema, supporting sales trend reporting and regulatory dashboards
- Applied Great Expectations for automated data validation, detecting anomalies, schema mismatches, and missing values in ingest pipelines.
- Enabled data security through AWS Lake Formation and RBAC within Snowflake, aligning with GDPR and internal compliance frameworks.
- Worked closely with the data science team to prepare time-series and categorical datasets used for trial modeling and segmentation scoring.
- Migrated historical datasets using legacy tools like Hive, HDFS, and Sqoop to cloud-native Snowflake tables during early-stage modernization.
- Maintained data dictionaries, lineage diagrams, and transformation logic in Confluence, collaborating with business and compliance stakeholders.

Environment: AWS Glue, Apache NiFi, EMR, PySpark, dbt, Snowflake, Airflow, Great Expectations, Hive, HDFS, Sqoop, Lake Formation, Spark SQL, Confluence, SQL, Informatica, HBase, Sqoop, AWS for ELT, Python for Data Cleaning and Scripting, Apache Kafka, Amazon Kinesis, Data Protection and Compliance, data quality management tools.

TECHNICAL SKILLS

Data Engineering Tools:	Apache Spark, MapReduce, Hive, Databricks, Data Modeling, Snowflake, Data
warehouse:	Snowflake, Azure Data Factory, Redshift
ETL Tools:	Talend, Informatica, Apache NiFi
Database platforms:	Teradata, HBase, Cassandra, Oracle Database, MySQL, MongoDB, SSIS, SQL Server Reporting Services (SSRS), PostgreSQL.
Microsoft Azure:	Azure Data Factory, Azure VM, Azure Databricks, Azure Data Lake Storage, Azure Hubs, Azure Synapse Analytics, Azure HDInsight, Azure Blob Storage.
AWS:	AWS, S3, EC2, Lambda, Glue, AWS Lake Formation, Kinesis Data Firehose, EMR
Programming Languages:	Python, R, Structured Query Language (SQL), PySpark, Scala, Java, Spark-SQL, UNIX
Python lib:	Pandas, SciPy, Scikit-learn, Seaborn, TensorFlow, Beautiful Soup, NumPy
Data Visualization:	Power BI, Tableau, Excel, Matplotlib, Looker, Excel.
GCP:	BigQuery, Google Storage.
Software & Model Deployment:	Git, Agile methodologies, Jira, Scrum, Docker, Kubernetes.

EDUCATION

Master’s in Information Technology | University of Cumberland | **Williamsburg, KY.**
Bachelor’s in Management Information System | University of Texas at Arlington | **Arlington, TX.**